

Villanova Off-Campus Housing Analysis

Estimating rent, supply, and affordability from a stratified cluster sample

104

ELIGIBLE LISTINGS

40

LISTINGS SAMPLED

146

UNITS OBSERVED

1,000

BOOTSTRAP ITERATIONS

Khanh (Ada) Nguyen

R | Survey Sampling | Bootstrap Inference | Tableau

Shared housing defines the affordable market

At the \$1,500 threshold, 90% of affordable observed units were shared.

RENT GAP

\$717

private minus shared estimate

AFFORDABLE AT \$1,500

54%

of 146 observed units

AFFORDABLE AT \$1,200

27%

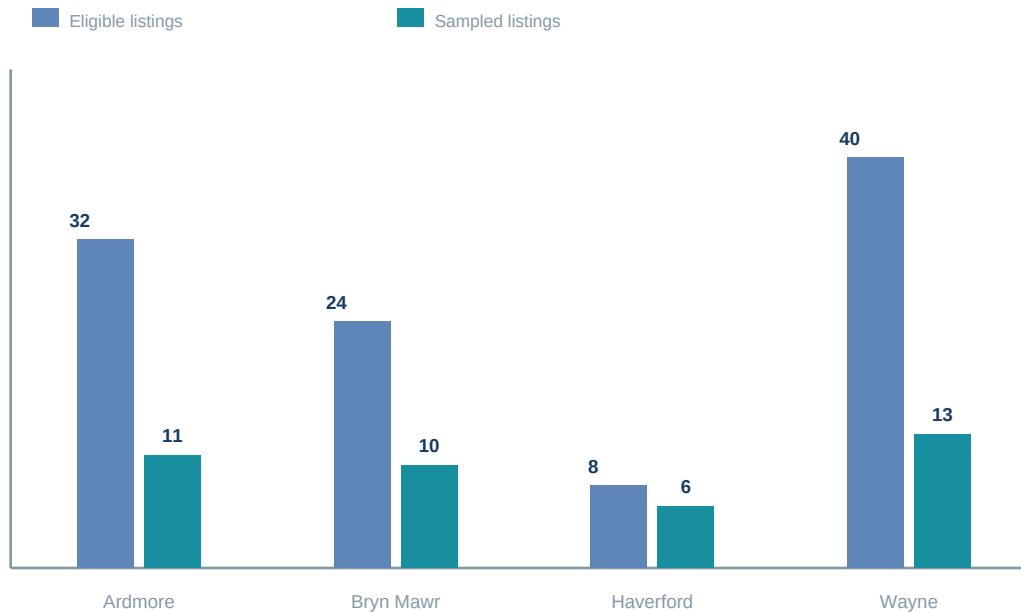
nearly all shared

DECISION CONTEXT

Sharing reduced the estimated monthly rent per person from \$1,950 to \$1,233.

The result helps students compare the cost tradeoff between privacy and affordability while recognizing uncertainty from a moderate cluster sample.

A practical probability sample replaced a full census



DESIGN

Stratified one-stage cluster sampling

Strata

Four nearby communities

Clusters

Rental listings

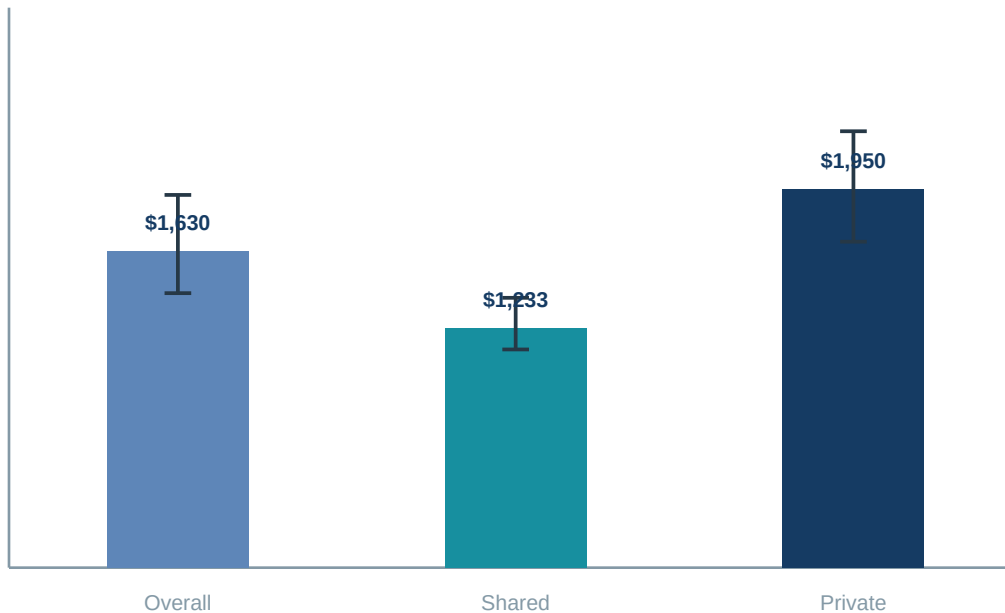
Elements

Units within listings

Allocation combined a five-listing minimum per stratum with proportional allocation of the remaining sample.

Khanh (Ada) Nguyen

Private housing carried a substantial premium



DESIGN-BASED ESTIMATES

\$1,233

shared per person

\$1,950

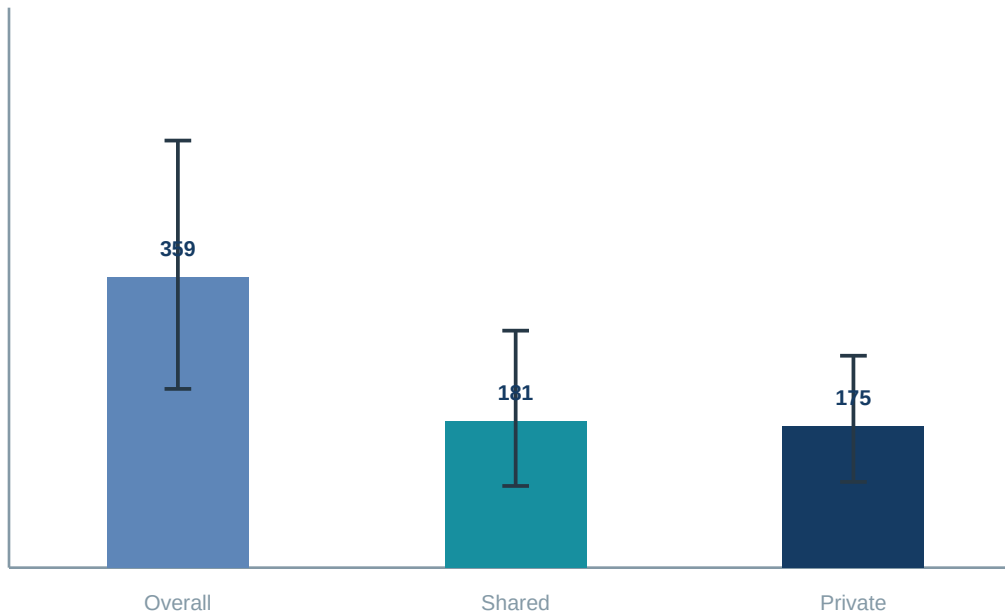
private per person

\$717

estimated monthly gap

Bootstrap intervals show limited overlap.

Rental supply estimates carried substantial uncertainty



ESTIMATED SUPPLY

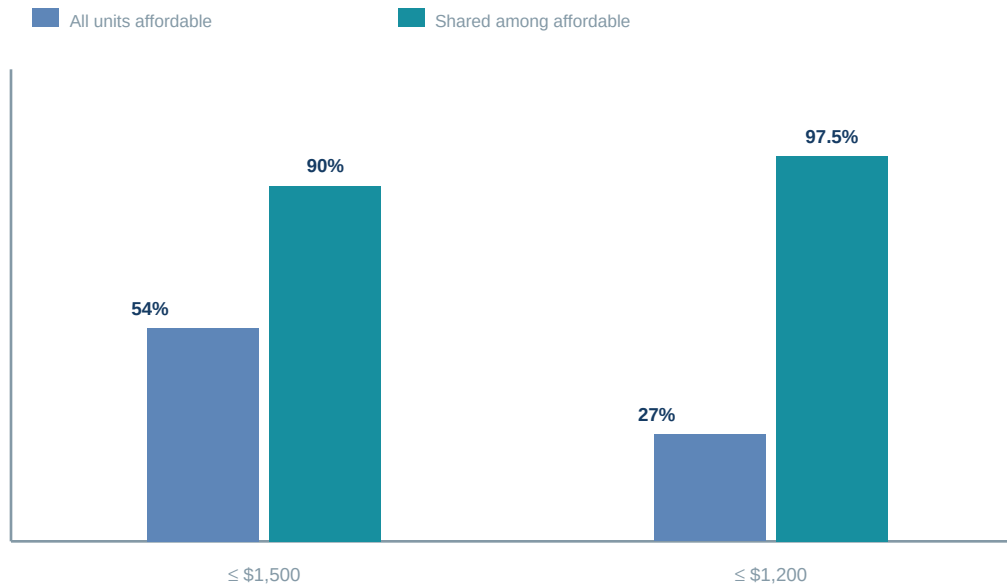
359 units

95% CI: 221 to 528

Shared and private totals were similar.

Wide intervals reflect the 40-listing sample and variation in the number of units per listing.

A tighter budget sharply reduced available options



AT \$1,500

54%

of observed units

90% shared

AT \$1,200

27%

of observed units

97.5% shared

The dashboard turns the sample into a housing decision tool

Tableau dashboard

Student Housing Market Near Villanova University

Compare rent, observed supply, housing type, and potential savings across Ardmore, Bryn Mawr, Haverford, and Wayne.

A reproducible workflow with clearly bounded claims

METHOD

- Ratio estimation within strata
- Population-weighted estimates
- Cluster bootstrap with $B = 1,000$

DATA QUALITY

- Eligibility rules applied consistently
- Duplicate and short-term listings excluded
- Every unit collected in sampled clusters

DELIVERABLES

- Reproducible R analysis
- Documented housing sample
- Interactive Tableau dashboard

LIMITATIONS

Zillow-only listings provide a snapshot rather than a permanent census. The frame excludes off-platform rentals, houses, townhomes, and short-term listings. Haverford had a small stratum, and unequal cluster sizes widened uncertainty for population totals.

PORTFOLIO TAKEAWAY

Sampling design made an incomplete housing market measurable and decision-ready.